ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO5 – Case Study Based Assignment Report

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Executive Summary

Question	Finding
Does the as-built link meet 3 dB margin?	**No.** As-built loss 22.52 dB (design 21.25 dB, +1.27 dB). Rx power −21.52 dBm; margin 0.38 dB (Ch 3) and −1.74 dB (Ch 5) after penalties.
What limits the link?	**Power and penalties, not noise.** OSNR = 47 dB vs 20 dB needed (27 dB spare). PMD 0.77 ps ≪ 10 ps. Dispersion penalty 1.00 dB.
Dominant excess loss	Event 3 (LC/UPC patch-panel connector, 1.20 dB, −22 dB): +0.70 dB. Event 2 (mechanical splice, 0.62 dB): +0.57 dB.
Channel 5 root cause	**Channel-specific transmitter:** ER 4.5 dB gives 3.22 dB penalty. The ER ratio predicts Q = 4.66 vs measured 4.7. Fibre is shared and cannot be the cause.
Quantum-limit gap	Ideal direct detection: 26.9 photons/bit = −44.6 dBm. Receiver sensitivity (−24 dBm) is **20.6 dB** from the limit.
Selected upgrade	**Option R: 4 × 400G Si-photonic, 75 GHz**: 300 GHz, 5.33 b/s/Hz, 37.5 pJ/bit, OSNR margin 21 dB.
Deployment decision	**NO for the as-built link; YES for the optimised design** once repairs pass verification (see Final Engineering Decision).

Stage 1 – Interpretation of the Measurement / Performance Requirement

1.1 Instruments and the quantity each validates

Instrument	Quantity validated	Use on this link
Optical power meter + stabilised source	Absolute power, end-to-end insertion loss	Gives as-built loss and received power; checks margin against −24 dBm
Optical spectrum analyser (OSA)	Channel wavelength/frequency, per-channel power, OSNR (0.1 nm), ASE	Confirms G.694.1 alignment and that OSNR is not limiting
OTDR (1550 nm)	Fibre length, attenuation, event position, event loss, reflectance	Locates event 2 (31.50 km) and events 3/4 (59.90 km)
Eye / BER tester	Q-factor, extinction ratio, jitter, eye width, BER	Separates Ch 5 transmitter fault from shared-fibre fault
CD / PMD analyser	Dispersion D, differential group delay	Verifies 17 ps/(nm·km) and PMD tolerance before 10G/coherent use

1.2 Loss, penalty and noise components for a practical budget

- **Loss:** fibre attenuation (0.21 dB/km × 60 km), connectors, fusion and mechanical splices, demultiplexer (6.0 dB), plus repair margin.
- **Penalties:** chromatic-dispersion penalty, extinction-ratio (ER) penalty, PMD penalty, reflection (interferometric) noise, and nonlinear penalty if launch power is raised.
- **Noise:** EDFA ASE (OSNR), receiver thermal and shot noise (quantum limit), relative-intensity and timing jitter.

1.3 Eye opening, jitter, ER, Q-factor and BER

A partially closed eye means a smaller vertical opening (lower Q) and a narrower horizontal opening (higher total jitter). Low ER means the "0" level carries too much power, which shrinks the signal swing and costs receiver sensitivity. Q relates to BER by $BER = \frac{1}{2}\,\mathrm{erfc}(Q/\sqrt{2})$. The 10^{-12} target requires **$Q \geq 7.03$**. Ch 3 (Q = 7.6) passes; Ch 5 (Q = 4.7) fails by a wide margin.

1.4 OTDR limits on this link

Dynamic range (DR) must exceed the one-way trace loss plus 6 dB or the far end disappears into noise. The **event dead zone (EDZ)** is the minimum separation at which two reflective events are seen as separate. The **attenuation dead zone (ADZ)** is the distance after a reflection before the loss

of the next event can be measured. Longer pulses raise DR but enlarge both dead zones, so DR and resolution conflict. Events 3 and 4 are only 9 m apart, which makes this a direct test of the trade-off.

1.5 Standards and connector / splice choices

- **ITU-T G.652.D:** low-water-peak single-mode fibre, zero dispersion near 1310 nm, D ≈ 17 ps/(nm·km) at 1550 nm; the standard fibre assumed here.
- **ITU-T G.694.1:** DWDM frequency grid anchored at 193.1 THz with 12.5, 25, 50, 100 GHz spacing and flexible grid (12.5 GHz slot width granularity).
- **Splices:** fusion (≈ 0.02–0.1 dB, negligible reflectance) versus mechanical (≈ 0.1–0.5 dB, reflectance −30 to −40 dB).
- **Connectors:** SC (2.5 mm ferrule, large), LC (1.25 mm, high density), MPO (multi-fibre parallel), CS (compact duplex for high-density pluggables); UPC versus APC (8° angled, lower reflectance).

1.6 Practical constraints

No fibre re-laying; system margin ≥ 3 dB; fixed C-band spectrum; and energy per bit must fall as capacity rises 20-fold (80 Gb/s to 1.6 Tb/s).

Performance Requirement Statement

The repaired and upgraded 60 km G.652.D link shall achieve: (1) total loss ≤ 21.25 dB with every OTDR event within allowance (connector ≤ 0.5 dB, fusion splice ≤ 0.05 dB, reflectance ≤ −50 dB); (2) system margin ≥ 3 dB on every channel after dispersion and ER penalties; (3) OSNR ≥ required value of the selected format; (4) Q ≥ 7.03 and BER ≤ 10^{-12}, ER ≥ 8.2 dB; (5) PMD ≤ 10 ps (10 % of the 100 ps bit period); (6) 1.6 Tb/s on the G.694.1 grid on the existing fibre pair at the lowest practical energy per bit.

Stage 2 – Link Budget, Dispersion and OSNR

2.1 As-built link loss versus design

Fibre attenuation = 60 km × 0.21 dB/km = **12.60 dB**. The as-built value replaces design allowances by measured values: four connectors are event 3 (1.20 dB) plus three measured at 0.5 dB (including event 4); splices are 12 fusion at 0.05 dB (including event 1) plus mechanical event 2 at 0.62 dB.

Component	**Design (dB)**	**As-built (dB)**	**Excess (dB)**
Fibre 60 km × 0.21	12.60	12.60	0.00
Connectors	4 × 0.5 = 2.00	1.20 + 3 × 0.5 = 2.70	+0.70
Splices	13 × 0.05 = 0.65	12 × 0.05 + 0.62 = 1.22	+0.57
Demultiplexer	6.00	6.00	0.00
Total	**21.25**	**22.52**	**+1.27**

2.2 Received power, penalties and margin

Dispersion penalty: $B \cdot D \cdot L \cdot \sigma\lambda = 10e9 \times 17e\text{-}12 \times 60 \times 0.03 = 0.306$

$\delta = 5 \log[1 + 2\pi(0.306)^2] = 5 \log(1.588) = 1.00$ dB

ER penalty: $PP = 10 \log[(r+1)/(r-1)]$, $r = 10^{(ER/10)}$

Ch 3: r = 7.943 → 1.10 dB Ch 5: r = 2.818 → 3.22 dB

Received power (as-built): +1 − 22.52 = −21.52 dBm

PMD = 0.1 × √60 = 0.775 ps (tolerance 10 % × 100 ps = 10 ps)

Quantity	**Channel 3 (ER 9.0 dB)**	**Channel 5 (ER 4.5 dB)**
Launch power (dBm)	+1.00	+1.00
As-built loss (dB)	22.52	22.52
Received power (dBm)	−21.52	−21.52
Dispersion penalty (dB)	1.00	1.00
ER penalty (dB)	1.10	3.22
Effective received power (dBm)	−23.62	−25.74
Receiver sensitivity (dBm)	−24.00	−24.00
System margin (dB)	**0.38**	**−1.74**
Required margin / shortfall (dB)	3.00 / 2.62	3.00 / 4.74

Before penalties the margin is 2.48 dB, which is already below 3 dB, consistent with the power-meter observation. Note: the −24 dBm sensitivity is quoted for ER ≥ 8.2 dB; I apply the absolute ER penalty to both channels as a conservative approach, and it correctly exposes Ch 5.

2.3 OSNR and PMD

OSNR = 58 + P_in − NF = 58 + (−6) − 5 = 47 dB (0.1 nm)

Margin over requirement: 47 − 20 = 27 dB

Noise is not limiting. A booster-only link has no downstream amplifier, so the receiver is **power- and penalty-limited**. PMD (0.775 ps = 0.8 % of the bit period) is negligible, and dispersion (σ = 30.6 ps, 0.31 UI) costs only 1.0 dB.

2.4 Corrective action to reach 3 dB margin

The dominant excess loss is event 3 (+0.70 dB), followed by event 2 (+0.57 dB). Even a perfect repair leaves margin of only 1.65 dB for Ch 3, so launch power must also rise. Required effective power ≥ −21 dBm, so minimum launch = −21 + 1.00 + 1.10 + 21.25 = **+2.35 dBm**; I adopt **+3 dBm per channel** (total +12 dBm for eight channels).

Scenario	Loss (dB)	Ch 3 margin (dB)	Ch 5 margin (dB)
As-built, +1 dBm	22.52	0.38	−1.74
Repair event 3/4 only, +1 dBm	21.82	1.08	−1.04
Repair events 2 and 3/4, +1 dBm	21.25	1.65	−0.47
Repair both + launch +3 dBm	21.25	3.65	1.53
Plus Ch 5 transmitter fixed (ER ≥ 9 dB)	21.25	3.65	3.65

Nonlinear check: +3 dBm per channel on eight channels at 100 GHz spacing over 60 km is below typical FWM/XPM/SBS limits for G.652, but this must be confirmed on the OSA and BER tester after the change.

DESIGN DECISION 1 – Link-budget configuration

I recommend repairing events 3/4 first, then event 2, then raising launch power to +3 dBm per channel and replacing the Ch 5 transmitter because the as-built link fails the 3 dB requirement (margin 0.38 dB, Ch 5 −1.74 dB) through excess loss (+1.27 dB) and penalties (2.10 dB for Ch 3), while OSNR (47 dB) and PMD (0.77 ps) have large reserves. The corrected budget gives a 3.65 dB margin.

Stage 3 – OTDR, Splice and Connector Strategy

3.1 Pulse-width selection

The OTDR sees the fibre plant (the demultiplexer is not in the trace). One-way trace loss = 22.52 − 6.00 = **16.52 dB**, so required DR = 16.52 + 6 = **22.52 dB**. Spatial pulse length = (c/n)·τ with c/n = $2.998\times10^{8}/1.4682 = 2.042\times10^{8}$ m/s.

Pulse	DR (dB)	EDZ (m)	ADZ (m)	Spatial length	DR ≥ 22.52 dB?	Events 3/4 (9 m apart)
10 ns	20	1	4	2.04 m	**No** (−2.5 dB)	Separate; event 4 loss measurable, but far end is lost
100 ns	**27**	**5**	**15**	**20.4 m**	**Yes (+4.5 dB)**	**Separate** (EDZ 5 < 9 m); event 4 loss **not** measurable (ADZ 15 > 9 m)
1 µs	33	40	100	204 m	Yes (+10.5 dB)	Merged (EDZ 40 > 9 m)

Selected: 100 ns for the full link. It is the shortest pulse with adequate DR, and it still separates events 3 and 4. The 20.4 m spatial length gives a theoretical half-pulse resolution of 10.2 m, so the 9 m separation is marginal and I confirm it with a supplementary short pulse. The loss of event 4 cannot be read at 100 ns because the ADZ (15 m) exceeds the 9 m cord; this matches the "loss not resolved" observation.

Test arrangement: launch cable and tail cable (≥ 150 m, longer than the 100 ns ADZ) so the first and last connectors are qualified; 1550 nm; IOR 1.4682; 1–3 min averaging; test **bidirectionally** and average to remove gain/loss artefacts; add a 10 ns shot from the receiver-side panel (where events 3 and 4 are near the OTDR) to resolve event 4 loss (ADZ 4 m < 9 m), and confirm with the power meter.

3.2 Event interpretation

Event	Position	Loss	Reflect.	Interpretation	Excess vs allowance
1	12.00 km	0.05 dB	none	Good fusion splice; within allowance	0
2	31.50 km	0.62 dB	−34 dB	**Mechanical splice** (index-matching gel gives the reflectance; loss from cleave angle, core offset, aged gel)	+0.57 dB
3	59.90 km	1.20 dB	−22 dB	**Poor LC/UPC patch-panel connector**: contaminated or damaged end face, or air gap; a healthy UPC is ≤ −50 dB	+0.70 dB
4	59.909 km	n/r	−40 dB	Second connector of a 9 m patch cord; reflectance acceptable; power meter gives 0.5 dB	0 (within allowance)

Property	Fusion splice	Mechanical splice
Typical loss	0.02–0.1 dB (here 0.05)	0.1–0.5 dB (here 0.62)
Reflectance	Essentially none	−30 to −40 dB (here −34 dB)
Stability	Permanent, ageing-free	Gel and alignment drift with temperature/time
Suitability for DWDM backbone	**Preferred**	Emergency / temporary only

3.3 Round-trip delay and location uncertainty

Round-trip delay $t = 2nL/c$; location error $\Delta L = L \times \Delta n/n$ with $\Delta n = \pm 0.002$, so $\Delta L/L = \pm 0.136$ %.

Event	Distance (km)	Round-trip delay (µs)	Location uncertainty (± m)
1	12.000	117.5	16.3
2	31.500	308.5	42.9
3	59.900	586.7	81.6
4	59.909	586.8	81.6

The ±43 m uncertainty at event 2 defines the search zone for the splice closure: use route records and sheath markers, and set the correct group index from the cable data sheet to shrink this. Events 3 and 4 are inside the terminating building, so the 82 m uncertainty is resolved by visual identification of the patch panel.

3.4 Connector selection for the patch-panel repair

Type	Characteristics	Verdict
SC	2.5 mm ferrule, push-pull, large footprint	Unnecessarily bulky for modern panels
LC	1.25 mm ferrule, duplex, high density, widely used on DWDM mux/transceivers	**Selected**
MPO	12–24 fibres in parallel	Not needed for a single fibre pair
CS	Very compact, for dense pluggable ports	Consider only if the pluggable demands it

Choice: LC/APC with factory-polished, fusion-spliced pigtails. The 8° angle sends the reflected light into the cladding, giving reflectance ≤ −60 dB versus ≈ −50 dB for UPC (the failed connector shows −22 dB). Insertion loss for a good connector is ≤ 0.3 dB, inside the 0.5 dB allowance. APC must mate only with APC; use APC/UPC hybrid cords at any UPC equipment port. Clean and inspect every end face (IEC 61300-3-35) before mating.

3.5 Repair ranking and verification

Rank	Event	Justification
1	**Events 3/4** (patch panel)	Largest excess loss (+0.70 dB), worst reflectance (−22 dB), easy access (clean or re-terminate in the building), and it is next to the receiver. Reflections also disturb the booster and interferometric noise.
2	**Event 2** (mechanical splice, 31.50 km)	+0.57 dB excess and a long-term reliability risk. Needs closure access (±43 m search) and re-fusion. Both repairs are needed: repairing event 3 alone gives only 1.08 dB margin at +1 dBm.
–	Event 1	Within allowance; no action.

Verification: (i) repeat bidirectional OTDR at 100 ns: event 2 ≤ 0.1 dB non-reflective; event 3 ≤ 0.5 dB with reflectance ≤ −60 dB (APC) and event 4 ≤ −50 dB; (ii) 10 ns shot confirms event 4 loss; (iii) power meter and source give end-to-end loss ≤ 21.25 dB (fibre plant ≤ 15.25 dB without the demultiplexer).

DESIGN DECISION 2 – OTDR fault-localization configuration

I recommend giving events 3/4 the highest priority: clean and replace the LC/UPC connector with an LC/APC fusion-spliced pigtail, then re-fuse event 2, testing at 100 ns bidirectionally with launch/tail cables plus a 10 ns reverse shot because event 3 has the largest excess loss (+0.70 dB) and worst reflectance (−22 dB), 100 ns is the only setting that gives both DR (27 dB ≥ 22.52 dB) and a 5 m EDZ that separates the 9 m-spaced events, and the 10 ns shot (ADZ 4 m) is needed to measure event 4 loss.

Stage 4 – Eye Diagram, BER and Quantum Limit

4.1 BER from Q-factor

Channel	Q	BER = ½ erfc(Q/√2)	vs 10^{-12} target	Status
Channel 3	7.6	1.5×10^{-14}	Better (×68)	Pass
Channel 5	4.7	1.3×10^{-6}	Worse (×1.3 million)	**Fail**
Target	7.03	1.0×10^{-12}	–	–

4.2 Jitter and eye width

$$TJ = DJ + 14.07 \cdot RJ$$

Ch 3: $0.18 + 14.07 \times 0.012 = 0.349$ UI $\rightarrow$ eye width $= 1 - 0.349 = 0.651$ UI

Ch 5: $0.30 + 14.07 \times 0.022 = 0.610$ UI $\rightarrow$ eye width $= 1 - 0.610 = 0.390$ UI

Horizontal (timing) margin: Ch 5 retains only 0.39 UI versus 0.65 UI, so clock recovery has little tolerance. **Vertical (noise) margin:** the optical power margin to the Q = 7.03 target is $10 \log(7.6/7.03) = +0.34$ dB for Ch 3 and $10 \log(7.03/4.7) =$ **−1.75 dB** for Ch 5. The Ch 3 value agrees with the budget margin (0.38 dB), which validates the budget.

4.3 Is Channel 5's low Q explained by its ER?

Item	Channel 3	Channel 5
ER (dB) / linear r	9.0 / 7.943	4.5 / 2.818
ER penalty (dB)	1.10	3.22
Eye-swing factor (r−1)/(r+1)	0.776	0.476
Predicted Q for Ch 5 = 7.6 × 0.476/0.776	–	**4.66** (measured 4.7)
Differential ER penalty	–	2.12 dB

The ER swing ratio reproduces the measured Q to within 1 %. Both channels share the same fibre, loss (22.52 dB), received power and dispersion penalty (1.0 dB), so none of these can create a Q difference. The root cause is channel-specific: the Ch 5 transmitter (laser bias too high or modulator drive/swing too low, or ageing). The higher DJ (0.30 vs 0.18 UI) and RJ (0.022 vs 0.012 UI) support this, as a low-ER, bias-limited laser shows pattern-dependent turn-on delay.

Recovery evidence: ER ≥ 8.2 dB (target ≥ 9 dB), Q ≥ 7.03 (expected ≈ 7.6 at the same power), TJ ≤ 0.35 UI, eye width ≥ 0.65 UI and BER ≤ 10^{-12}. With the repairs and +3 dBm the effective power rises 3.27 dB, so by linear Q scaling Q rises to ≈ 16 (idealised; the real value will be lower but far above 7.03).

4.4 Quantum limit for direct detection

BER = ½ exp(−N) = 10^{-12} → N = ln(5 × 10^{11}) = 26.94 ≈ 27 photons per 1-bit

Photon energy at 1550 nm: hν = hc/λ = 6.626e-34 × 2.998e8 / 1550e-9 = 1.282e-19 J

Minimum power: P = N·hν·B = 26.94 × 1.282e-19 × 10e9 = 34.5 nW = −44.6 dBm

Gap to receiver sensitivity: −24 − (−44.6) = 20.6 dB

If power is averaged over equiprobable 1s and 0s the limit is 3 dB lower (−47.6 dBm), a 23.6 dB gap. The 20 dB gap is typical of a PIN + transimpedance receiver dominated by thermal noise, with ER and non-ideal quantum efficiency adding more.

4.5 Direct detection versus coherent PSK

For BER = 10^{-12}, ½ erfc(x) = 10^{-12} gives x = Q/√2 with Q = 7.034. Homodyne PSK: √(2N) = Q/√2 → N = $Q^2/4$. Heterodyne PSK: √N = Q/√2 → N = $Q^2/2$.

Scheme	N (photons/bit)	Power at 10 Gb/s (nW)	Sensitivity (dBm)	vs direct (dB)	Gap from −24 dBm (dB)
Direct detection	26.9	34.5	−44.6	0	20.6
Homodyne PSK	12.4	15.9	−48.0	−3.4	24.0
Heterodyne PSK	24.7	31.7	−45.0	−0.4	21.0

Homodyne detection is 3 dB better than heterodyne (half the photons), because heterodyne spreads signal power over the image band. Ideal homodyne is 3.4 dB better than direct detection.

4.6 Implications for the coherent upgrade

- A strong local oscillator makes coherent receivers **shot-noise limited**, so they approach the quantum limit regardless of receiver thermal noise. This is why coherent detection gives the higher sensitivity needed for QAM.
- The ideal limit scales with rate: about −38 dBm at 100G, −35 dBm at 200G and −32 dBm at 400G (homodyne, indicative). The repaired link delivers about −18 to −21 dBm per channel,

more than 10 dB above these limits, so the upgrade is limited by ASE/OSNR and implementation penalties (LO phase noise, ADC, DSP), not photon starvation.

- Higher-order formats need more photons per bit and higher OSNR (QPSK 12.5 dB, 16QAM 19.5 dB, 400G 26 dB), so margin must be validated per option, as done in Stage 5.

DESIGN DECISION 3 – Eye / BER performance configuration

I recommend rejecting Channel 5 as-is and replacing (or re-biasing) its transmitter to reach ER ≥ 9 dB, with acceptance on Q ≥ 7.03, TJ ≤ 0.35 UI, eye width ≥ 0.65 UI and BER ≤ 10^{-12} because its BER is 1.3×10^{-6} (Q = 4.7), its eye is only 0.39 UI wide, and its 3.22 dB ER penalty reproduces the Q deficit (predicted 4.66), proving a channel-specific transmitter fault rather than a fibre problem. The receiver is 20.6 dB above the quantum limit, so no receiver-side remedy is needed.

Stage 5 – Integrated WDM Upgrade and Energy Strategy

5.1 G.694.1 channel plan

f(n) = 193.1 THz + n × 0.05 THz (50 GHz grid)

n = +3: f = 193.1 + 0.15 = 193.25 THz

λ = c/f = 2.998e8 / 193.25e12 = 1551.36 nm

Option Q occupied width per channel = 32 GBd × (1 + 0.1) = 35.2 GHz

slots m = ceil(35.2 / 12.5) = ceil(2.82) = 3 → 37.5 GHz per channel

8 channels × 3 slots = 24 slots = 300 GHz (centre frequencies on a 6.25 GHz grid)

5.2 Spectrum, spectral efficiency and OSNR margin

The repaired-link OSNR is 47 dB (booster-limited, 58 + P_in − NF).

Option	Channels	Slot	Occupied (GHz)	SE (b/s/Hz)	Required OSNR (dB)	OSNR margin (dB)
P	16 × 100G DP-QPSK	50 GHz fixed	800	2.00	12.5	34.5
Q	8 × 200G DP-16QAM	3 × 12.5 GHz	300	5.33	19.5	27.5
R	4 × 400G Si-photonic	75 GHz fixed	300	5.33	26.0	21.0
Current	8 × 10G NRZ	100 GHz	800	0.10	20	27

5.3 Module power and energy per bit

Option	Modules	W/module	Total power (W)	pJ/bit	Saving vs legacy
Legacy 10G SFP+ (for 1.6 Tb/s)	160	3.5	560	350	–
P: 16 × 100G	16	22	352	220	37 %
Q: 8 × 200G	8	20	160	100	71 %
R: 4 × 400G Si-photonic	4	15	60	37.5	89 %

Energy per bit = W / (bit rate): 3.5/10G = 350; 22/100G = 220; 20/200G = 100; 15/400G = 37.5 pJ/bit

The legacy route is also infeasible: 160 channels on a 100 GHz grid need 16 THz, far more than the C-band (≈ 4.4–4.8 THz). **Si-photonic integration** puts modulators, germanium photodetectors, waveguides and couplers on CMOS-compatible wafers; this cuts drive voltage, coupling and packaging loss and cost, and lets a single laser, driver and DSP overhead be shared across 400 Gb/s, which is why energy per bit falls by 6× against P.

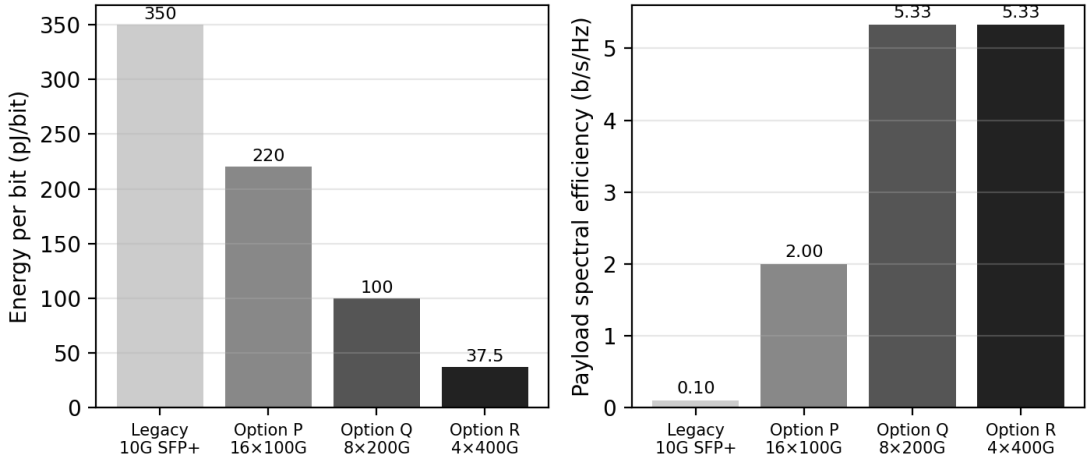


Figure 1 – Energy per bit and payload spectral efficiency of the options.

5.4 Sequence of repair, verification and upgrade

1. **Re-characterise** with OTDR (100 ns bidirectional + 10 ns) and power meter to freeze the baseline.
2. **Repair events 3/4** (clean, then LC/APC fusion pigtails); re-test. This is the cheapest and largest gain (+0.70 dB).
3. **Re-fuse event 2** at the closure; re-test.
4. **Replace the Ch 5 transmitter** and raise launch to +3 dBm; verify Q, ER, TJ, BER on the existing 10G system. This proves the fibre plant before new technology is added.
5. **Install Option R** (4 × 400G) and retire the 10G channels to avoid cross-modulation between NRZ and coherent signals.
6. **Re-verify** OTDR, OSA OSNR, pre-FEC BER, then a 24–72 h soak test.

Repairing before upgrading ensures that any later fault is attributable to the new equipment, not to the unresolved connector and splice, which would otherwise be hidden by FEC.

5.5 Ranking and selection

Criterion	Best	Middle	Worst
Spectrum (GHz)	Q = R (300)	–	P (800)
OSNR margin (dB)	P (34.5)	Q (27.5)	R (21.0), still ≥ 3 dB
Energy (pJ/bit)	R (37.5)	Q (100)	P (220)
Overall	**R**	Q	P

Selected: Option R. It ties Q for the lowest spectrum (300 GHz, leaving ~94 % of the C-band free), has the lowest energy (37.5 pJ/bit, −83 % vs P) and the fewest modules (4). Its 21 dB OSNR margin is the lowest but far above any practical requirement. A 75 GHz slot equals six 12.5 GHz slots on the G.694.1 flex grid. Total launched power is only +9 dBm at +3 dBm/channel, leaving room to raise per-channel power if the pluggable datasheet needs a higher receive power. Option Q is the fallback if more reach margin is later needed.

Measurements to repeat after the upgrade: bidirectional OTDR baseline; power-meter loss and per-channel power; OSA centre frequency (193.1 THz grid), OSNR at 0.1 nm; pre-FEC BER and Q, post-FEC error-free operation; CD and PMD (DGD); module power draw; 24–72 h BER soak.

DESIGN DECISION 4 – Integrated optimization strategy

I recommend repair → verify on 10G → upgrade to Option R (4 × 400G Si-photonic, 75 GHz) → re-verify because the repaired link delivers a 3.65 dB margin and 47 dB OSNR (21 dB above the 26 dB need for 400G), PMD and dispersion are negligible, and Option R gives 1.6 Tb/s in 300 GHz (5.33 b/s/Hz) at 37.5 pJ/bit, which is 89 % less energy than the legacy module approach.

Stage 6 – Verification and Critical Evaluation

Requirement	Target	As-found	After repair / upgrade (predicted)
Total optical link loss	≤ 21.25 dB	22.52 dB (+1.27)	**21.25 dB** (events 2, 3 restored to 0.05 / 0.5 dB; ≈ 21.05 dB with 0.3 dB LC/APC)
Received optical power	≥ −24 dBm + 3 dB	−21.52 dBm; effective −23.62 / −25.74 dBm	−18.25 dBm; effective −20.35 dBm (≥ −21 dBm)
System margin	≥ 3 dB	Ch 3: 0.38 dB; Ch 5: −1.74 dB	**3.65 dB** both channels

OTDR event condition	No event above allowance	Ev 2: 0.62 dB; Ev 3: 1.20 dB, −22 dB	Ev 2 ≤ 0.1 dB; Ev 3 ≤ 0.5 dB, ≤ −60 dB; Ev 4 ≤ −50 dB
Eye / Q / jitter / ER	Q ≥ 7.03; ER ≥ 8.2 dB	Ch 3: Q 7.6, ER 9.0, TJ 0.349 UI; Ch 5: Q 4.7, ER 4.5, TJ 0.610 UI	Ch 5 ER ≥ 9 dB, Q ≥ 7.6 (≈ 16 at +3 dBm), TJ ≈ 0.35 UI
BER and quantum-limit gap	BER ≤ 10^{-12}	Ch 3: 1.5 × 10^{-14}; Ch 5: 1.3 × 10^{-6}; gap 20.6 dB	BER ≪ 10^{-12} on both channels; gap 20.6 dB (receiver) / 26 dB (Rx power)
Upgrade spectrum and energy	1.6 Tb/s, min pJ/bit	80 Gb/s, 800 GHz, 350 pJ/bit	1.6 Tb/s, 300 GHz, 5.33 b/s/Hz, **37.5 pJ/bit**

Consistency check. (1) The budget margin for Ch 3 (0.38 dB) matches the Q-based margin (0.34 dB), so the loss accounting and the measured Q are consistent. (2) The two OTDR excess losses sum exactly to the 1.27 dB budget excess. (3) The ER swing ratio reproduces the measured Ch 5 Q. (4) OSNR and PMD have large reserves, so the faults are loss and transmitter faults, as diagnosed. (5) All repair-stage values are predictions; they become evidence only after the post-repair measurements are logged, which is why the verification gate precedes the upgrade.

Final System Design

Workflow flowchart

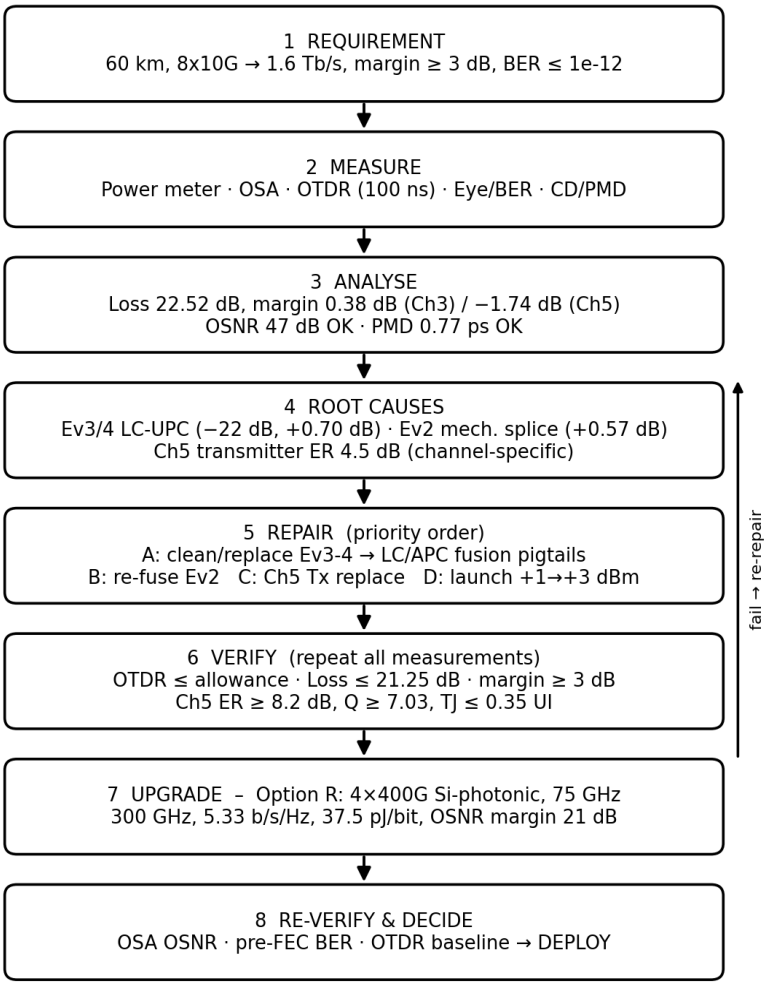


Figure 2 – Diagnostic, repair and upgrade workflow.

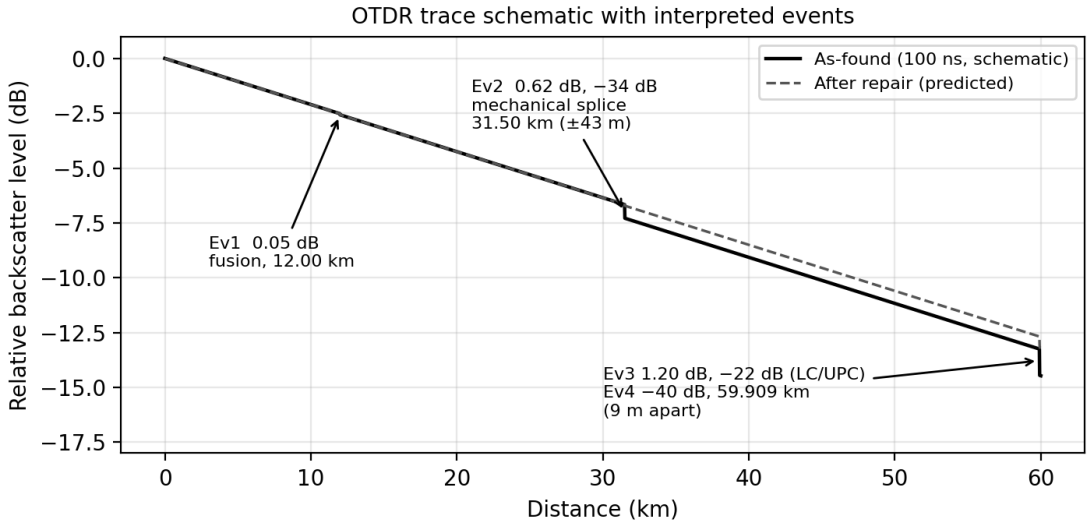


Figure 3 – Interpreted OTDR events, as-found versus predicted after repair (schematic).

Final parameter and diagnostic table

Parameter	Value	Decision adopted
Fibre loss	12.60 dB	No action
As-built / design loss	22.52 / 21.25 dB	Repair to ≤ 21.25 dB
Dispersion penalty; PMD	1.00 dB; 0.775 ps (≤ 10 ps)	No compensation needed
OSNR (booster)	47 dB (need 20 dB)	Not limiting
Launch power	+1 dBm → +3 dBm/ch	Raise; verify nonlinearity
OTDR setting	100 ns, bidirectional, launch/tail cables; 10 ns reverse shot	Adopted
Priority 1 / 2	Event 3/4 connector / event 2 splice	LC/APC fusion pigtail / re-fuse
Ch 5 transmitter	ER 4.5 dB, Q 4.7, BER 1.3 × 10^{-6}	Replace or re-bias; ER ≥ 9 dB
Quantum limit	−44.6 dBm direct; −48.0 homodyne; −45.0 heterodyne	Receiver 20.6 dB above limit
Selected upgrade	Option R, 4 × 400G, 75 GHz, 37.5 pJ/bit	300 GHz, 21 dB OSNR margin
Ch n = +3 (50 GHz grid)	193.25 THz, 1551.36 nm	G.694.1 compliant

Effect of each change (predicted)

Change	Why	Predicted effect
LC/APC pigtail at event 3/4	+0.70 dB excess; −22 dB reflectance	Margin 0.38 → 1.08 dB; reflectance ≤ −60 dB
Re-fuse event 2	+0.57 dB excess; reflective mechanical splice	Margin → 1.65 dB
Launch +1 → +3 dBm	Penalties exceed repaired margin	Margin → 3.65 dB
Replace Ch 5 transmitter	ER 4.5 dB gives 2.12 dB extra penalty	Ch 5 margin 1.53 → 3.65 dB; Q 4.7 → ≥ 7.6; BER 1.3×10^{-6} → ≤ 10^{-14}
Option R coherent upgrade	Capacity ×20 within 300 GHz	1.6 Tb/s; 350 → 37.5 pJ/bit (−89 %)

Final Engineering Decision

- **Budget and margin:** repaired loss 21.25 dB; +3 dBm launch gives −18.25 dBm received, effective −20.35 dBm, margin 3.65 dB ≥ 3 dB. Excess loss (+1.27 dB) is traced entirely to events 2 and 3, with event 3 dominant.
- **OTDR:** 100 ns satisfies DR (27 ≥ 22.52 dB) and the EDZ (5 m < 9 m); the ADZ limit is handled with a 10 ns reverse shot. Priority is events 3/4, then 2; LC/APC and fusion splicing remove the reflective defects.
- **Eye and BER:** Ch 3 is healthy (Q 7.6, BER 1.5×10^{-14}). Ch 5 failure is a transmitter ER fault (predicted Q 4.66 vs 4.7); replacing it restores Q ≥ 7.6 and TJ ≈ 0.35 UI.
- **Quantum limit and coherent suitability:** the receiver is 20.6 dB from the limit; coherent detection is shot-noise limited and received power is > 10 dB above the coherent limit, so ASE (47 dB OSNR, 21 dB margin for Option R) is the governing limit.
- **Spectrum and energy:** G.694.1 plan gives 1.6 Tb/s in 300 GHz (5.33 b/s/Hz) at 37.5 pJ/bit, 6× better than Option P and 9× better than the 10G modules.

Conditions and risk control. Deployment is released only if post-repair measurements show: loss ≤ 21.25 dB, every event within allowance, margin ≥ 3 dB, Ch 5 ER ≥ 8.2 dB with Q ≥ 7.03, and no nonlinear penalty at +3 dBm. If any gate fails, the contingency is: raise per-channel power further add a pre-amplifier or lower-loss demultiplexer, or fall back to Option Q (27.5 dB OSNR margin, same 300 GHz). These steps deliver 1.6 Tb/s on the existing fibre pair with no re-laying.